MIL-C-52681A(ME)
28 August 1975
SUPERSEDING
MIL-C-52681(ME)
16 March 1970

MILITARY SPECIFICATION

CAMERA SECTION, TOPOGRAPHIC REPRODUCTION SET, TRUCK MOUNTED, PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for preservation, packaging, packing, and marking of a Camera Section, Topographic Reproduction Set, Truck Mounted (SC 3610-97-CL-E13).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.

Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-R-3593	- Refrigeration and Cooling Equipment (Excluding Household Refrigerators) Packaging of.

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MIL-C-3615 - Camera, Copying, Lithographic, Packaging of.

MIL-G-28554 - Generator Sets, Mobile Electric Power; Packaging of.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Preproduction pack. The supplier shall furnish a preproduction pack within the time frame specified (see 6.2) to prove prior to starting production packaging and packing, that the applied preservation, packaging, packing, and marking comply with the requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.2 Materials. Materials shall be as specified herein and in applicable specifications and standards. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Materials shall be free of defects.

3.3 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

3.3.1.2 Methods of preservation. Methods of preservation indicated by symbols herein shall conform to the corresponding symbols in MIL-P-116.

3.3.1.3 Handtools. The hammer, pliers, rule, screwdriver, shears, measuring tape, and adjustable wrench shall be preserved and packaged in accordance with PPP-P-40, Level A. The preserved and packaged tools shall be placed in the toolbox furnished as a set component.

3.3.1.4 Truck/van. The components and accessories of the truck/van shall be preserved in accordance with MIL-STD-281, Level A, (mobile), and as follows.

3.3.1.4.1 Screwjacks. The screws of the jacks shall be coated with Type P-11 preservative and the screws fully retracted into the jacks. Exterior unpainted surfaces of the jacks shall be coated with Type P-1 or P-19 preservative. The jacks shall be secured in the stowage compartment on the truck/van body.

3.3.1.4.2 One-hundred-foot power cable, cable reel, and cover. The cable assembly shall be wound on the cable reel provided and secured to prevent the cable from unwinding. The reel cover shall then be installed and the reel secured in the brackets provided.

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3.3.1.4.3 Water hose assembly, reel, and reel cover. The 50-foot water hose assemblies **shall** be wound on the hose reel provided and the reel secured to prevent the hose from unwinding. The reel cover shall then be placed on the hose reel.

3.3.1.4.4 Heaters and air conditioner. Heaters shall be completely drained of fuel. Components of the air conditioner shall be preserved in accordance with applicable requirements of MIL-R-3593, Level A.

3.3.1.5 Generator set (MAP). The generator set, accessories, repair parts, tools, and publications shall be preserved, packaged, packed, and marked in accordance with MIL-G-28554, Level A.

3.3.1.6 Copying camera. The camera installed in place shall be preserved and packaged in accordance with MIL-C-3615, Level B, except that the Method IA-16 barrier shall not be required.

3.3.1.7 Acetic acid, screwcap bottles, cleaning compound, developer, fixing bath, hypoeliminator, potassium ferricyanide, sodium triosulfate, and xylene. These items shall not require additional packaging. Items shall be consolidated packaged as specified in 3.3.1.19.

3.3.1.8 Pipe to hose adapters, pipe bushings, hose couplings, pipe nipples, and pipe reducers. Items requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. Items of like description shall be preserved together Method IC-1.

3.3.1.9 Paint brush and scrub brush. The bristles or hair of each brush shall be protected by an envelope or fiber pad to prevent deformation. Each brush shall then be preserved Method IC-1.

3.3.1.10 Laboratory aprons, compass bar, film clips, beam compass, tape dispenser, erasers, hypoestimator, magnifier, exposure meter, pencils, spectacles, sponges, pressure-sensitive tape, thermometer, halftone screens, paper towel dispenser, and squeegee. These items shall be preserved Method IC-1. When practical items of like description shall be preserved together in quantities specified for each set.

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3.3.1.11 Eight-foot, 9-inch cable assembly. The terminal lugs of the four conductors shall be secured together with tape and the cap covers shall be installed on the plugs. The cable assembly shall then be coiled in a minimum coil, not less than 10 times the cable diameter and secured with cotton tape or twine.

3.3.1.12 Telephone cable, funnel, padlocks, trays, metal and plastic pails, toolbox, waste receptacles, and copper (bare) wire. These items shall not require unit packaging. Items shall be consolidated packaged as specified in 3.3.1.19.

3.3.1.13 Cotton cloth, blotting paper, lens paper, and plastic sheet. The cotton cloth shall be rolled. Each roll, dozen or box as applicable, shall be preserved Method IC-1.

3.3.1.14 Fire extinguishers and ground rods. These items shall be placed and secured in their mounting brackets.

3.3.1.15 Alining instrument, liquid graduate, hot plate, time stop, and interval timer. Each of these items shall be preserved Method III in containers conforming to PPP-B-636, W5c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

3.3.1.16 Fluorescent and incandescent lamps. Each lamp shall be placed in a single-face corrugated fiberboard sleeve. The fluorescent lamps shall be consolidated together and the incandescent lamps shall be consolidated together in boxes conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

3.3.1.17 Microscope. The microscope shall be preserved and packaged Method IIb. The case shall be considered the inner container. The outer container shall be a box conforming to PPP-B-636, W5c, style optional. Box closure for the outer container shall be in accordance with the appendix to the box specification, Method V.

3.3.1.18 Filters. Each filter shall be wrapped with lens tissue and overwrapped with corrugated paperboard cushioning. The filter shall then be placed in a fiberboard box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

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3.3.1.19 Consolidated packaging. Small components, packaged and unpackaged, shall be placed in storage cabinets or any other available space provided for storage within the van body. Other components for which space is not available shall be consolidated packaged in a minimum number of boxes conforming to PPP-B-601, Domestic Type, Style I. The size and configuration of the boxes shall be determined by the available space that will permit securing within the van body or truck cab. Strapping of consolidated containers shall not be required. The boxes shall be immobilized within the van body or truck cab using blocking, bracing, or strapping as required, applied in a manner that will not damage the items or set components to which the boxes are secured.

3.3.2 Level C. The components of the set including the truck/van shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination.

3.4 Packing. Packing shall be Level A or C, as specified (see 6.2).

3.4.1 Level A. The truck/van containing the set components, prepared for shipment as specified in 3.3, shall be shipped unboxed (mobile). Wheels shall be free to rotate. The packed generator set (MAP) shall be placed inside the van body in any available space and shall be secured to prevent movement within the van body. When space is not available for securing within the van body, the generator set shall be shipped separately from but at the same time as the truck/van.

3.4.2 Level C. The truck/van containing the set components, preserved and packaged as specified in 3.3, shall be prepared for shipment unboxed (mobile) in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

3.5 Marking. In addition to any special marking required by the contract or purchase order, packages, consolidated containers, and unboxed (mobile) truck/vans shall be marked in accordance with MIL-STD-129.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects listed in 4.4.4. Presence of one or more defects shall be cause for rejection of the preproduction pack.

4.4 Quality conformance inspection.

4.4.1 Inspection stages.

(a) The first stage shall include inspection of procedures, methods, materials, containers, and packaging marking prior to placing the packages into location in the van or prior to placing the packages into consolidated containers.
(b) The second stage shall include inspection of closure and marking of the consolidated containers, and the arrangement and blocking and bracing of the located items and blocking and bracing of consolidated containers in the van.
(c) The third stage shall include inspection of the marked mobile van as prepared for shipment.

4.4.2 Unit of product. For the purpose of inspection, all packages or packs prepared for any applicable inspection stage that will comprise a complete pack, shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level in Table I. Acceptable Quality Level (AQL) shall be 2.5 percent defective.

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Table I. Classification of defects

No.	Defect	Level A	Level C
101.	Materials, methods, and containers not as specified. Each incorrect material, method, or container shall constitute one defect (3.2, 3.3, 3.4).	X	-
102.	Threads of screwjacks not coated with preservative and screws not fully retracted as specified (3.3.1.4.1).	X	-
103.	Cable reel not secured to prevent cable from unwinding (3.3.1.4.2).	X	-
104.	Hose reel not secured to prevent hose from unwinding (3.3.1.4.3).	X	-
105.	Heaters not completely drained of fuel (3.3.1.4.4).	X	-
106.	Bristles or hair of brushes not protected as specified (3.3.1.9).	X	-
107.	Cable assembly not coiled and secured as specified (3.3.1.11).	X	-
108.	Fire extinguishers and ground rods not secured in their mounting brackets (3.3.1.14).	X	-
109.	Lamps not packaged as specified (3.3.1.16).	X	-
110.	Filters not preserved and packaged as specified (3.3.1.18).	X	-
111.	Consolidated packaging not as specified (3.3.1.19).	X	-
112.	Wheels not free to rotate (3.4.1).	X	-
113.	Marking illegible, incomplete, incorrect, or missing (3.5).	X	X

5. PREPARATION FOR DELIVERY

(This section not applicable to this specification.)

6. NOTES

6.1 Intended use. This specification is intended to be used for direct reference in contracts or purchase orders for the preservation, packaging, packing, and marking of a Camera Section, Topographic Reproduction Set, Truck Mounted (NSN 3610-00-598-5811) or (NSN 3610-00-926-1757) for MAP.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction pack (see 3.1).
(c) Level of preservation and packaging and level of packing required (see 3.3 and 3.4).
(d) Any special marking required (see 3.5).

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6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the set components in accordance with this specification.

Custodian:

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Preparing activity:

Army - ME

Project No. 3610-A267

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DOCUMENT IDENTIFIER AND TITLE MIL-C-52681A(ME) CAMERA SECTION, TOPOGRAPHIC REPRODUCTION SET, TRUCK MOUNTED, PACKAGING OF

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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